

COP 4934 – Summer 2026

Group 11

Meeting Notes

Date: June 9, 2026

Time: 4:00 pm

Location: Library (Room 370B)

1. Tolga has not gotten a response back from Lee County regarding the data request he sent on 5/28/2026. Phillip will call the county to ask how long these data requests typically take.
2. Phillip and Andrew will start running the data harvest script (Tolga will continue running it) to collect historical data from the Lee County database. Discuss options next week to run the script on more systems.
3. Database: Plan to keep current database under Tolga's account for project.
4. Discussed potential restrictions for database access and backup options. No decision made for potential restrictions. Backup appears to be sufficient with snapshots taken daily and stored for 14-days.
5. Listed the analyses we plan to implement for the project.
 - i. DBSCAN (Density-Based Spatial Clustering of Applications with Noise) for clustering
 - Three (3) levels
 - ii. KDE (Kernel Density Estimation) for heat map
 - Base: single bandwidth
 - Optional: multiple bandwidth; weighted
 - iii. Getis-Ord G_i^* (aka G-i-star) for statistical validation of hot spots
 - Need to determine weighted matrix approach
6. Discussed an outline of the design report; to be added to by team as topics/sections are identified. Topics identified during the meeting included:
 - i. Lee County API and data attribute summary
 - ii. Project procedure for collecting historical data from County's database
 - iii. Project procedure for daily data refresh from County's database
 - iv. Algorithm description of HDBSCAN, KDE, and G_i^*
7. Action Items:
 - i. **(Phillip and Andrew)** Run the data harvest script.
 - ii. **(Phillip)** Call Lee County to ask how long data request is expected to take.
 - iii. **(Phillip)** Start an outline document for the team to update throughout project. Will be used to keep track of topics and sections to include in the design report.
 - iv. **(Phillip)** Prepare document summarizing the traffic data available from the County's api.